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SECTION 3. Emerging Trends and Technologies in IT Application (5 MAY 2023)

# Recycling for Recycling: RoI Cropping by Recycling a Pre-trained Attention Mechanism for Accurate Classification of Recyclables

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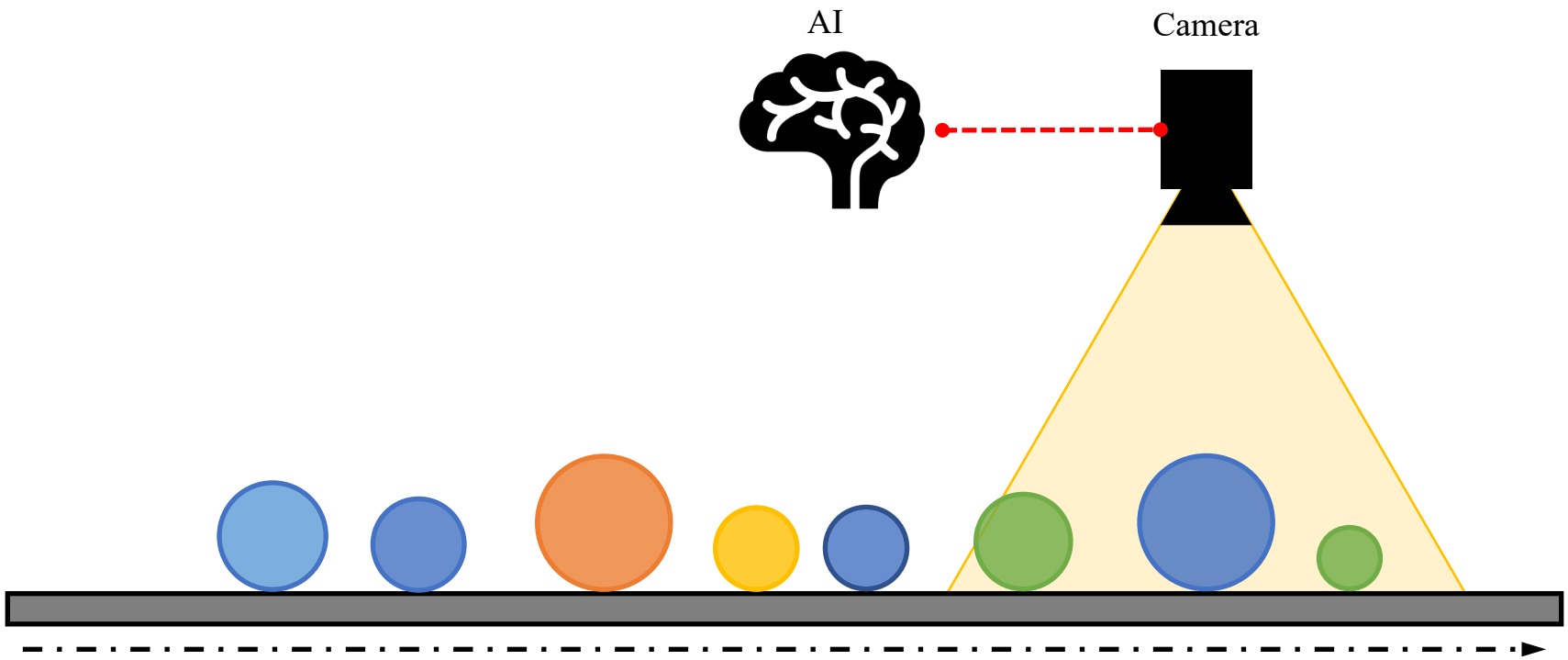
# Background

# Ideal System



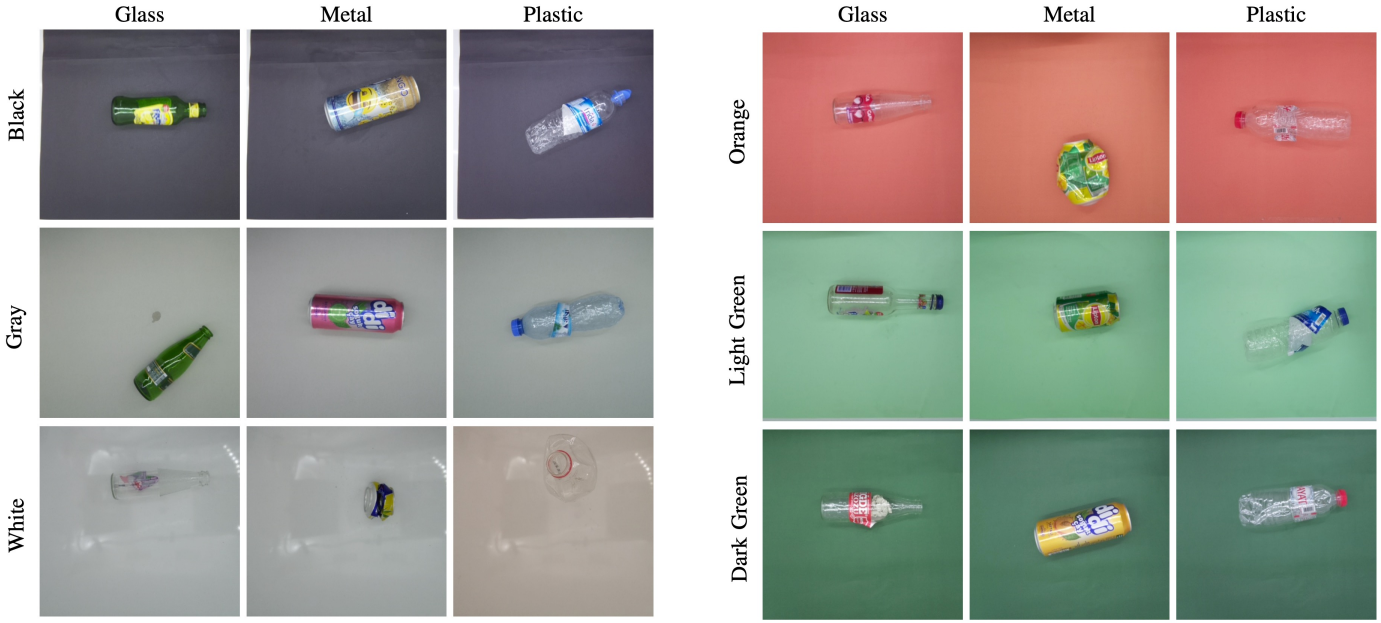

- Move recyclables **one by one** on a conveyor belt
- Classify each object via machine vision system
- Consider **end-to-end** object detection and recognition model
  - i. e. R-CNN * or YOLO **

* S. Ren, K. He, R. Girshick, and J Sun. "Faster R-CNN: Towards real-time object detection with region proposal networks." Advances in neural information processing systems. 2015.
** J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. "You only look once: Unified, real-time object detection." IEEE conference on computer vision and pattern recognition. 2016.

# Real-world Problem



Recyclable Solid Waste Dataset (RSWD) dataset *




**Limitations of an end-to-end model**

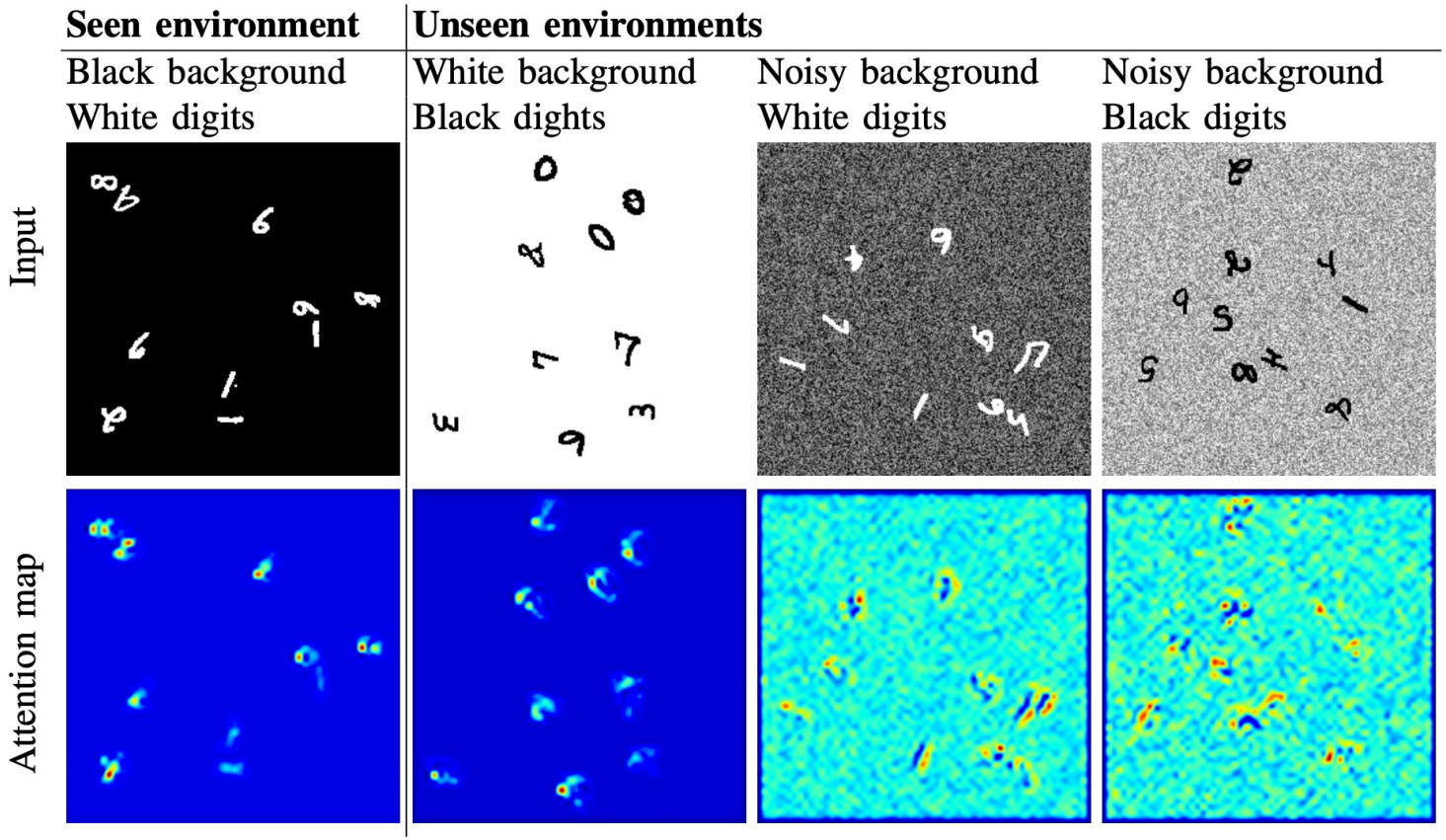
- Infinite kind of recyclable data type
  - i. e. colors, transparency, pose of recyclables, and various data collection environments.
- Practically impossible to construct dataset to train end-to-end models

**Build an object detection while minimizing the above**

- Should work universally in a variety of environments

*Recyclable Solid Waste Dataset, Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/hseyinsaidkoca/recyclable-solid- waste-dataset-on-5-background-co

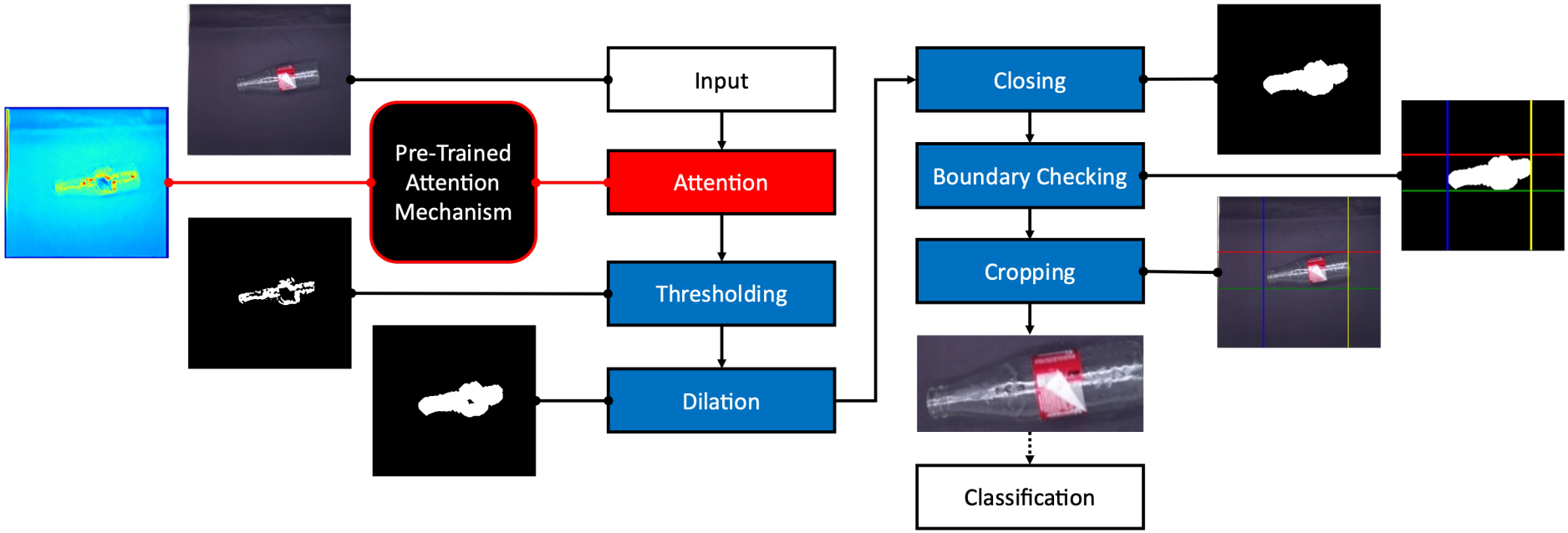
# Approach




- MNIST pre-trained attention properly emphasizes even for unseen environments
  - i. e. digit rotation or background color change
- Pre-trained attention can be utilized to detect recyclables

# Attention-based RoI Cropping






**Proposal**

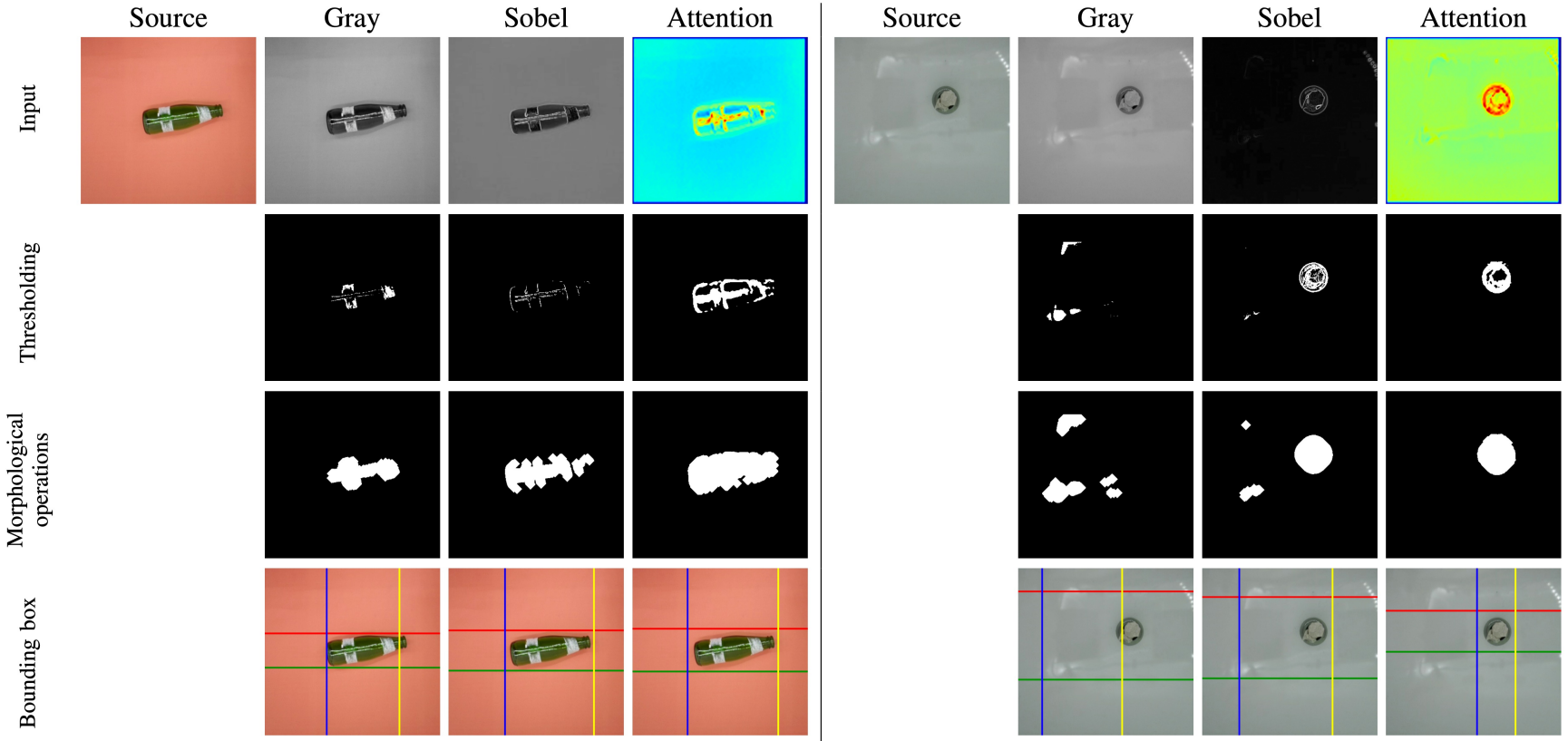
- RoI cropping method based on pre-trained attention

**Strength**

- Minimize the hindrance surrounding information for recyclables recognition
- No longer need an end-to-end object detection model

# RoI Cropping Results






- Comparison of RoI detection based on grayscaling, Sobel filtering, and attention mechanism
- For easy case (a), all three methods perform well
- In difficult case (b), an object on a transparent plate, only the attention method performs well

# Dataset



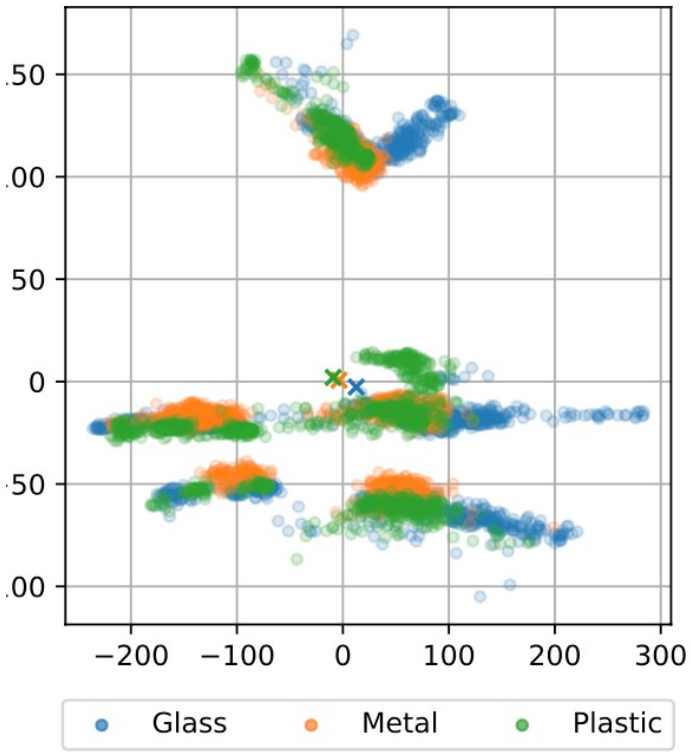
Recyclable Solid Waste Dataset (RSWD)*

| Class | Black | Gray | White | Orange | Light Green | Dark Green | Total |
|---|---|---|---|---|---|---|---|
| Glass | 364 | 351 | 92 | 348 | 345 | 161 | 1661 |
| Metal | 332 | 353 | 169 | 343 | 342 | 116 | 1655 |
| Plastic | 330 | 353 | 175 | 340 | 337 | 95 | 1630 |

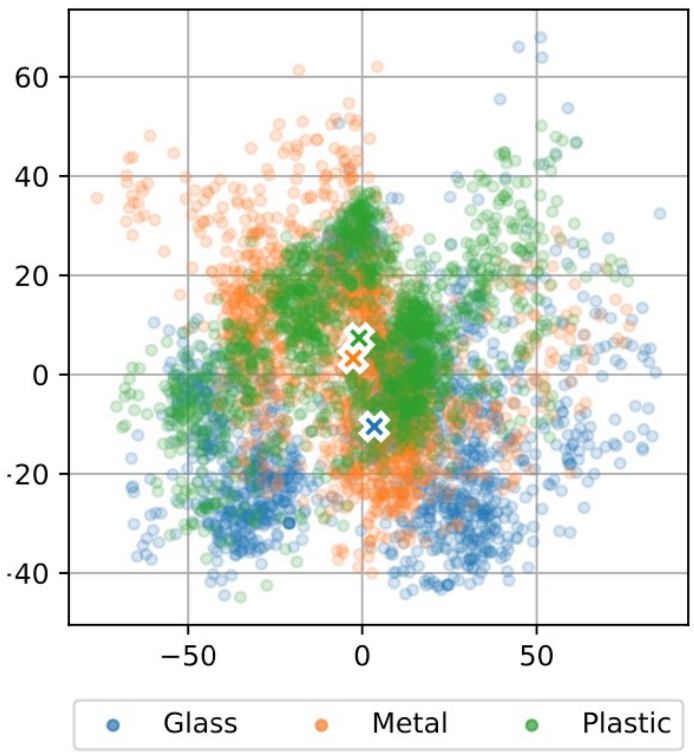
*Recyclable Solid Waste Dataset, Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/hseyinsaidkoca/recyclable-solid- waste-dataset-on-5-background-co

# Effect of RoI Cropping






F-ratio: 0.008




F-ratio: 0.220

- Before RoI cropping, 5 clusters according to the background color
- After RoI cropping, 3 clusters by each class, glass, metal, and plastic

# Classification Performance



| Model | ResNet50 * | | EffNetB0 ** | |
|---|---|---|---|---|
| RoI Crop. | ✗ | ✓ | ✗ | ✓ |
| Accuracy | 0.839 | **0.898** | 0.885 | **0.927** |
| Precision | 0.886 | **0.919** | 0.902 | **0.932** |
| Recall | 0.839 | **0.897** | 0.885 | **0.927** |
| F1-score | 0.828 | **0.893** | 0.869 | **0.927** |

- Widely adopted classifiers, ResNet50* and EfficientNet-B0**
- Performance improvement in all metrics on both classifiers

* He, Kaiming, et al. "Deep residual learning for image recognition." IEEE conference on computer vision and pattern recognition. 2016.
** M. Tan and Q. Le "EfficientNet: Rethinking model scaling for convolutional neural networks." International Conference on Machine Learning, vol. 97, 2019, pp. 6105–6114.

# Conclusions



**Reduce Hindrance**

- Pre-trained attention detects RoI accurately
- RoI cropping reduces interference by surrounding information

**Improve Efficiency/Performance**

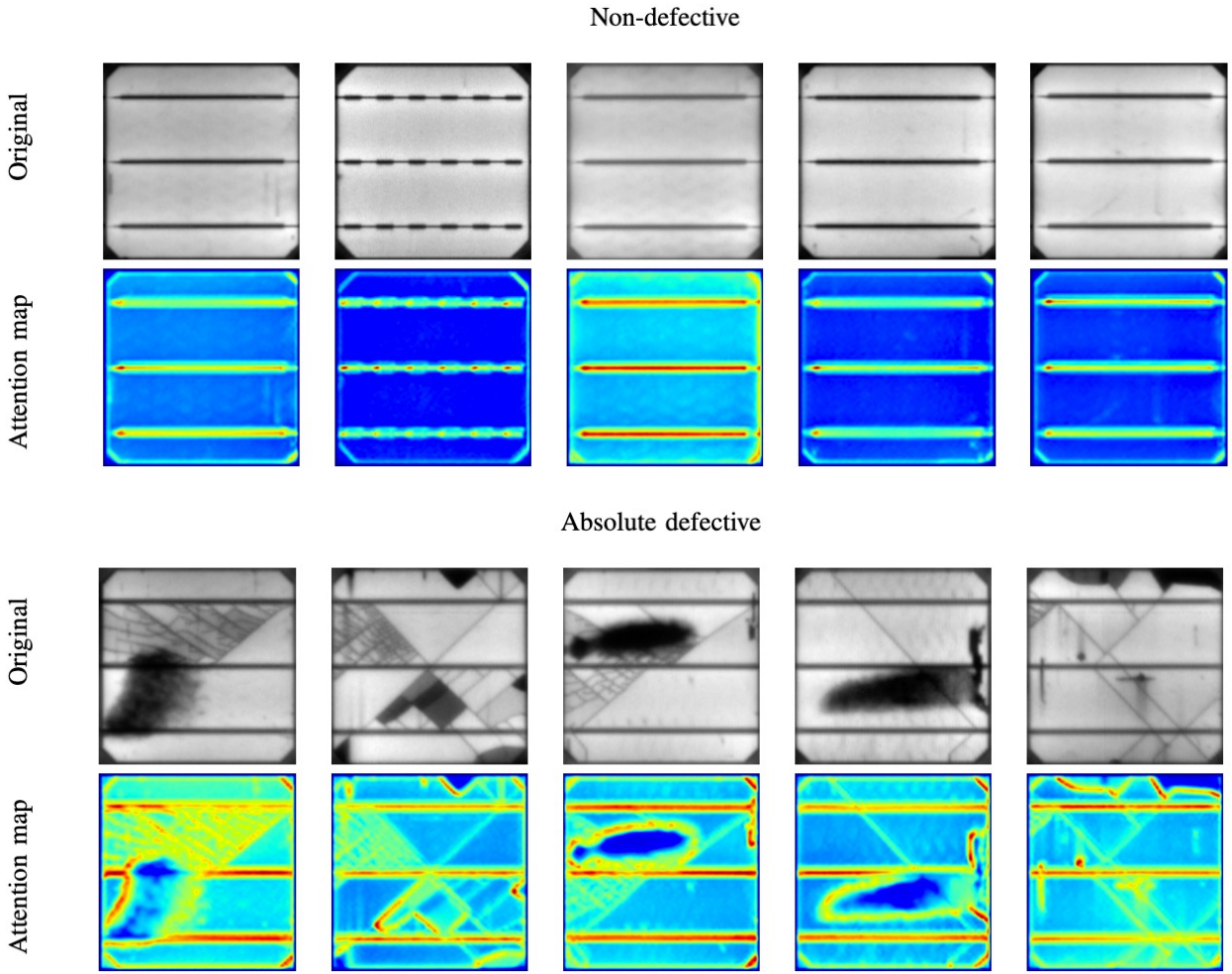
- RoI cropping reduces the image size and the computational burden of classifiers
- Classification performance is increased by focusing on object information

**Recycling for Recycling**

- Repurposing pre-trained attention contributes directly to resource recycling
- Data annotation and training end-to-end model is no longer needed

# Related Research






* YH. Park, M.J. Kim, U. Gim, and J. Yi. "Boost-up Efficiency of Defective Solar Panel Detection with Pre-trained Attention Recycling." *IEEE Transactions on Industry Applications*, 2023.

# Thank you